

11. Which of the following is NOT included in JRE?
 - a) JVM
 - b) Java class libraries
 - c) Compiler (javac)
 - d) Core classes
12. Which of the following is platform dependent?
 - a) Bytecode
 - b) Source code
 - c) JVM implementation
 - d) JDK version
13. The javac command generates:
 - a) Executable file
 - b) Class file
 - c) Jar file
 - d) Binary file
14. Why is the main method declared as static?
 - a) To allow JVM to call it without creating an object
 - b) To make code reusable
 - c) To restrict access from other classes
 - d) To ensure it runs faster
15. Which of the following main method declarations will successfully compile & run?
 - a) public static void main(String... args)
 - b) public static void main(String args[])
 - c) static public void main(String[] args)
 - d) All of the above
16. What happens if we declare main method as private?
 - a) Compile-time error
 - b) Runtime error - JVM cannot access it
 - c) Runs normally
 - d) Prints nothing
17. If a program has two main methods with different parameters:
 - a) Compilation fails
 - b) JVM always calls the main(String[] args)
 - c) JVM chooses randomly
 - d) JVM calls the overloaded one
18. Which of the following is a valid declaration?
 - a) int _value;
 - b) int \$value;
 - c) int value123;
 - d) All of the above
19. Which variable must be initialized at declaration?
 - A) Local variable
 - B) Instance variable
 - C) Static variable
 - D) Class variable
20. Command-line arguments are:
 - A) Compile-time inputs
 - B) Runtime inputs
 - C) JVM constants
 - D) Environment variables

1. Which of these naming conventions is correct?
 - a) Method → CalculateSalary()
 - b) Variable → EmployeeName
 - c) Class → EmployeeDetails
 - d) Constant → MaxSalary
2. Instance variables are created:
 - a) Once per class
 - b) Once per object
 - c) In static memory
 - d) During compilation
3. What is the default value of an uninitialized String instance variable?
 - a) Empty string ""
 - b) null
 - c) "null"
 - d) Compilation error
4. In java, if you execute program like : **java FileProcessor <filePath> <filename>** , what is the size of args?
 - a) 2
 - b) 3
 - c) 4
 - d) Undefined
5. In the same(above) program, value of args[2] is:
 - a) A
 - b) B
 - c) C
 - d) IndexOutOfBoundsException
6. Command line arguments are always:
 - a) Integers
 - b) Strings
 - c) Characters
 - d) Objects
7. To convert a command-line string argument "123" to integer, we use:
 - a) Integer.parseInt("123")
 - b) Integer.valueOf(123)
 - c) (int)"123"
 - d) toInt("123")
8. Static methods can access:
 - a) Only static data directly
 - b) Both static and non-static data directly
 - c) Only instance variables
 - d) None
9. Which of these cannot be declared static?
 - a) Methods
 - b) Variables
 - c) Constructors
 - d) Blocks
10. Scope of a local variable is limited to:
 - A) Class
 - B) Package
 - C) Method / Block
 - D) JVM

21. Which loads first into memory when class is loaded?

- a) Instance variables
- b) Static variables
- c) Local variables
- d) Constructor

22. What is output?

```
public class Test {  
    static int x;  
    public static void main(String[] args) {  
        System.out.println(x);  
    }  
}
```

- a) 0
- b) null
- c) Garbage value
- d) Compilation error

23. Which of these is not a primitive data type?

- a) int
- b) boolean
- c) String
- d) double

24. The default value of an uninitialized local when using variable is:

- a) null
- b) 0
- c) false
- d) Compiler error

25. Which will throw compile-time error?

- a) int x = 10;
- b) boolean flag = false;
- c) int x = null;
- d) String s = null;

26. Which is true about System.out.println?

- a) System is a class
- b) out is a static reference of PrintStream
- c) println is a method
- d) All of the above

27. If only JRE is installed (without JDK), what can you do?

- a) Compile and run Java code
- b) Only run Java bytecode
- c) Only compile Java code
- d) Neither compile nor run

28. Command-line arguments help to:

- A) Avoid writing main()
- B) Change code logic
- C) Avoid hard-coded values
- D) Improve JVM speed

29. What happens if a static variable is accessed using an object reference?

- a) Compile-time error
- b) Runtime error
- c) Allowed but discouraged
- d) JVM crash

30. Which is required for compilation of Java code?

- a) JVM only
- b) JRE only
- c) JDK
- d) None

31. Which statement about JVM is true?

- a) JVM is platform independent, bytecode is platform dependent
- b) Bytecode is platform independent, JVM is platform dependent
- c) Both JVM and bytecode are platform independent
- d) Both are platform dependent

32. What happens if you remove static from the main method?

- a) Program compiles and runs normally
- b) Program fails at runtime (JVM cannot call main)
- c) Compile-time error
- d) JVM automatically adds static

33. Which is true about String args in main method?

- a) JVM always passes them as primitive strings
- b) JVM passes them as array of String objects
- c) JVM passes them as Object[]
- d) JVM does not pass anything by default

34. What happens if you access arr[5] in an array of size 5?

- a) Returns 0
- b) Returns null
- c) ArrayIndexOutOfBoundsException
- d) Compilation error

35. Which of the following is true about arrays in Java?

- a) Arrays are objects stored in heap
- b) Arrays have fixed length once created
- c) Arrays can store primitive and objects both
- d) All of the above

36. Which of the following statements about local variables in Java is true?

- a) Local variables are declared inside a method and must be initialized before use
- b) Local variables can be accessed from any class
- c) Local variables are automatically initialized to default values
- d) Local variables are declared outside all methods

37. Which is a common use of command-line arguments?

- A) Declaring variables
- B) Passing file names or configurations
- C) Creating objects
- D) Handling exceptions